

Supplementary Material

Additional Model Diagnostics and Robustness Analyses

Data-generating process

Figure S1 illustrates the baseline type distribution. Figures S2–S3 describe the match-value function under that distribution. Within each realization, principal types and the matching-function objects c , A , and B are held fixed while ρ and δ vary. The match-value figures show the conditional value $E[q_{ij} | x_i, x_j] = Q + f(x_i, x_j)$, without match noise. They therefore isolate the structure principals and the broker must learn rather than variation from one set of noisy outcomes.

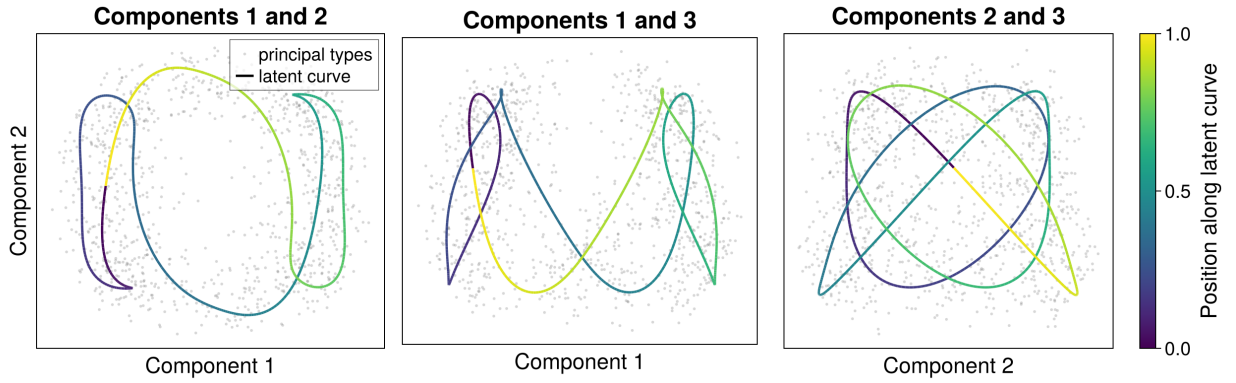


Figure S1: Three views of the baseline type distribution for all 1000 principals in realization 1. Principal types are drawn around a smooth latent curve and projected to the unit sphere in the model’s eight-dimensional type space. For visualization, the figure projects both the realized types and the noiseless curve onto the curve’s first three principal components. Gray points are realized principal types, the colored line is the latent curve, and color records position along that curve. The projection is used only for visualization; matching and learning use the complete eight-dimensional type vectors.

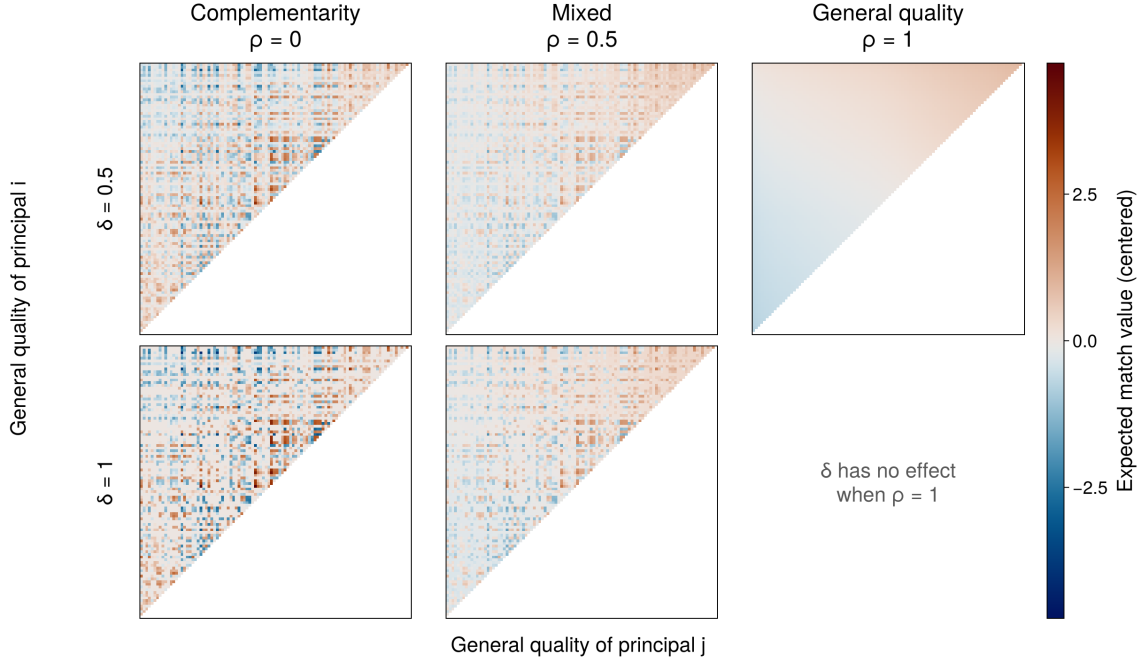


Figure S2: Centered conditional match value for 100 principal types selected at evenly spaced general-quality ranks from a 1000-principal realization (seed 1). Columns vary the composition of match value; rows compare baseline difficulty ($\delta = 0.5$) with high difficulty ($\delta = 1$). Because δ is inactive at $\rho = 1$, that boundary condition is shown once. The realized types and matching-function objects c , A , and B are identical across panels. General quality increases from left to right and from bottom to top. Each symmetric matrix is shown once in its upper triangle; the diagonal is excluded because principals cannot match with themselves. All panels use the same off-diagonal color scale. The matrices exclude match noise and are centered by subtracting their grand mean.

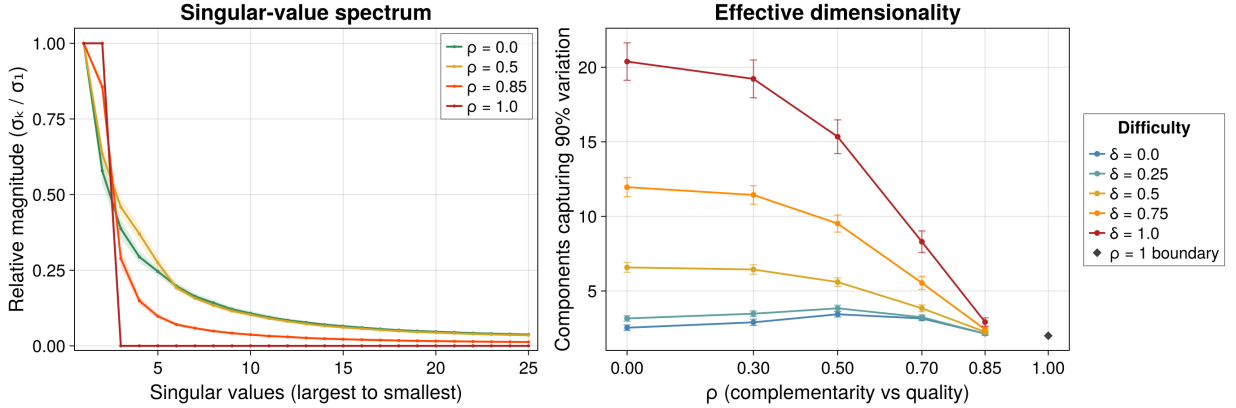


Figure S3: Dimensionality of the centered conditional match-value matrix over 50 independently generated sets of principal types and matching-function objects. Left: the first 25 singular values at $\delta = 0.5$. Each value is divided within its realization by the largest singular value, σ_1 . Lines are means across the 50 realizations and ribbons are pointwise 95% Monte Carlo intervals. Right: within each realization, the minimum number of singular components needed to capture 90% of match-value variation. Markers are means across the 50 realizations and whiskers are 95% Monte Carlo intervals. The pure-quality boundary $\rho = 1$, where δ is inactive, is shown once and unconnected.

Alternative structural measures

Figures S4–S6 reproduce the main structural analyses using the broker’s two other saved ego-network measures, Burt’s aggregate network constraint and Burt’s effective size. Both are recomputed for the broker every 20 periods. As in the main text, a regime is one scientifically distinct parameter configuration; duplicate grid coordinates and inactive parameters do not create additional regimes.

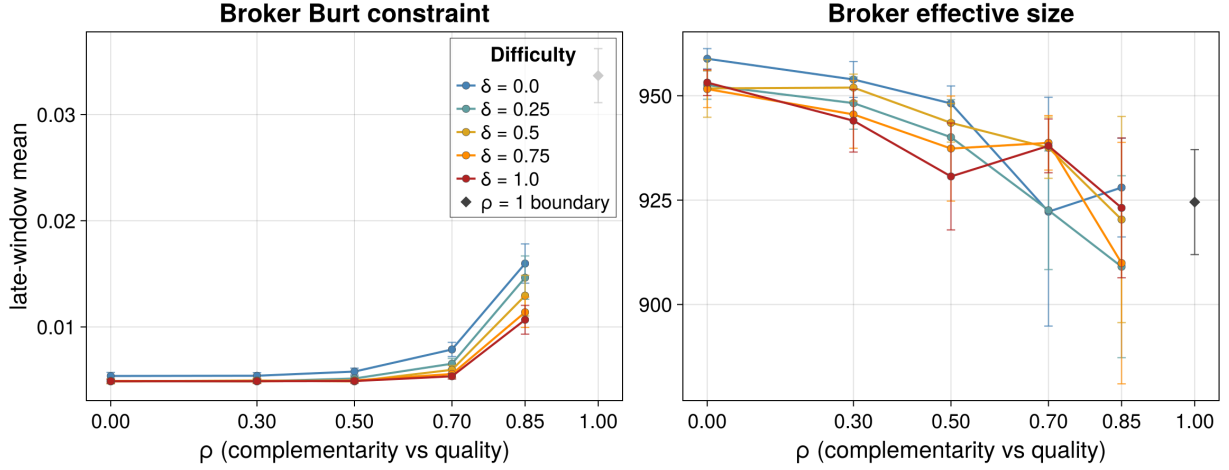


Figure S4: Broker Burt constraint (left) and broker effective size (right) across the $\rho \times \delta$ grid. Lines for each difficulty level δ span $\rho = 0$ through 0.85. The pure-quality boundary $\rho = 1$, where δ is inactive, is shown once and unconnected. Markers are late-window means over seeds and whiskers are 95% Monte Carlo intervals.

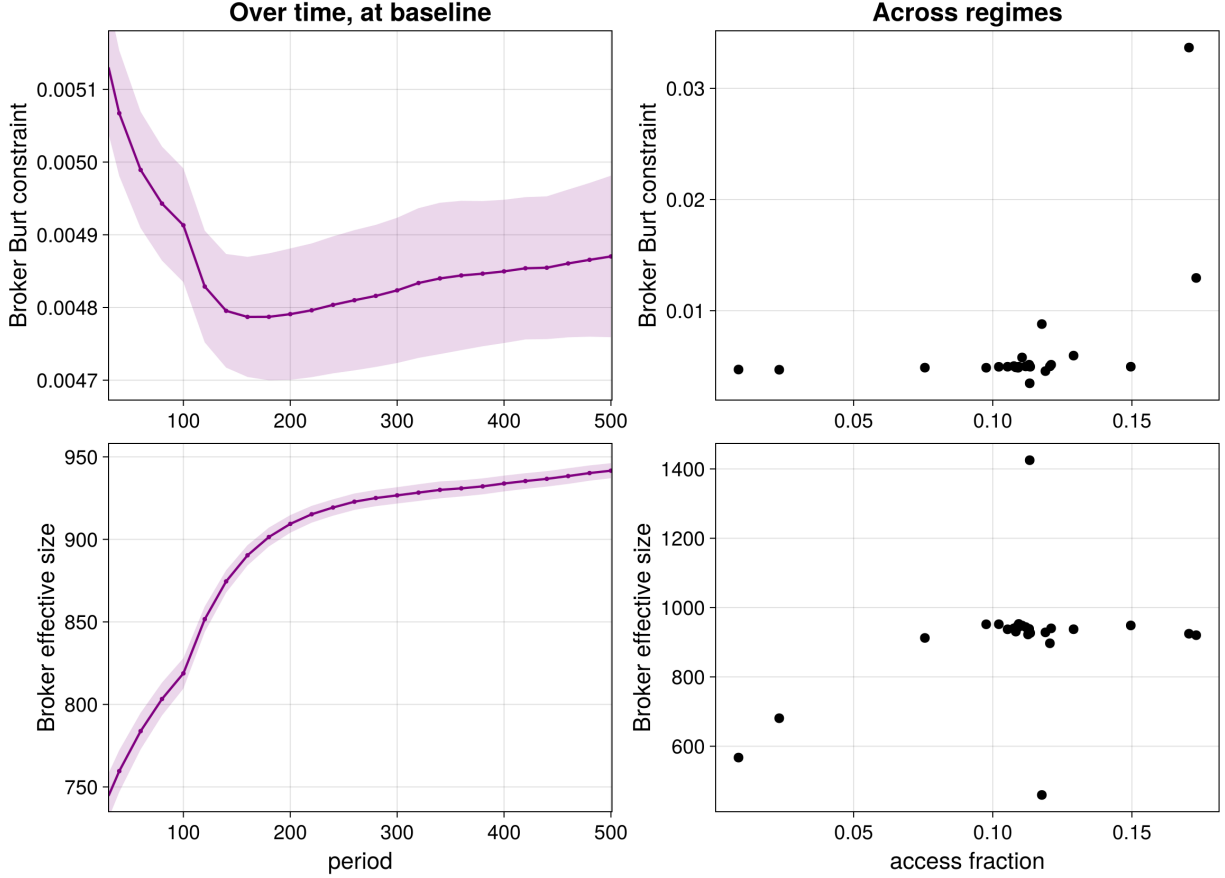


Figure S5: Broker Burt constraint (top) and broker effective size (bottom). Left: each measure over time at the baseline, shown as a 5-observation rolling ensemble mean over seeds with pointwise 95% Monte Carlo intervals. Network measures are computed every 20 periods. Right: each measure against the access fraction, with one late-window regime mean per one-at-a-time regime. Time starts at $t = 30$ to omit the initialization transient.

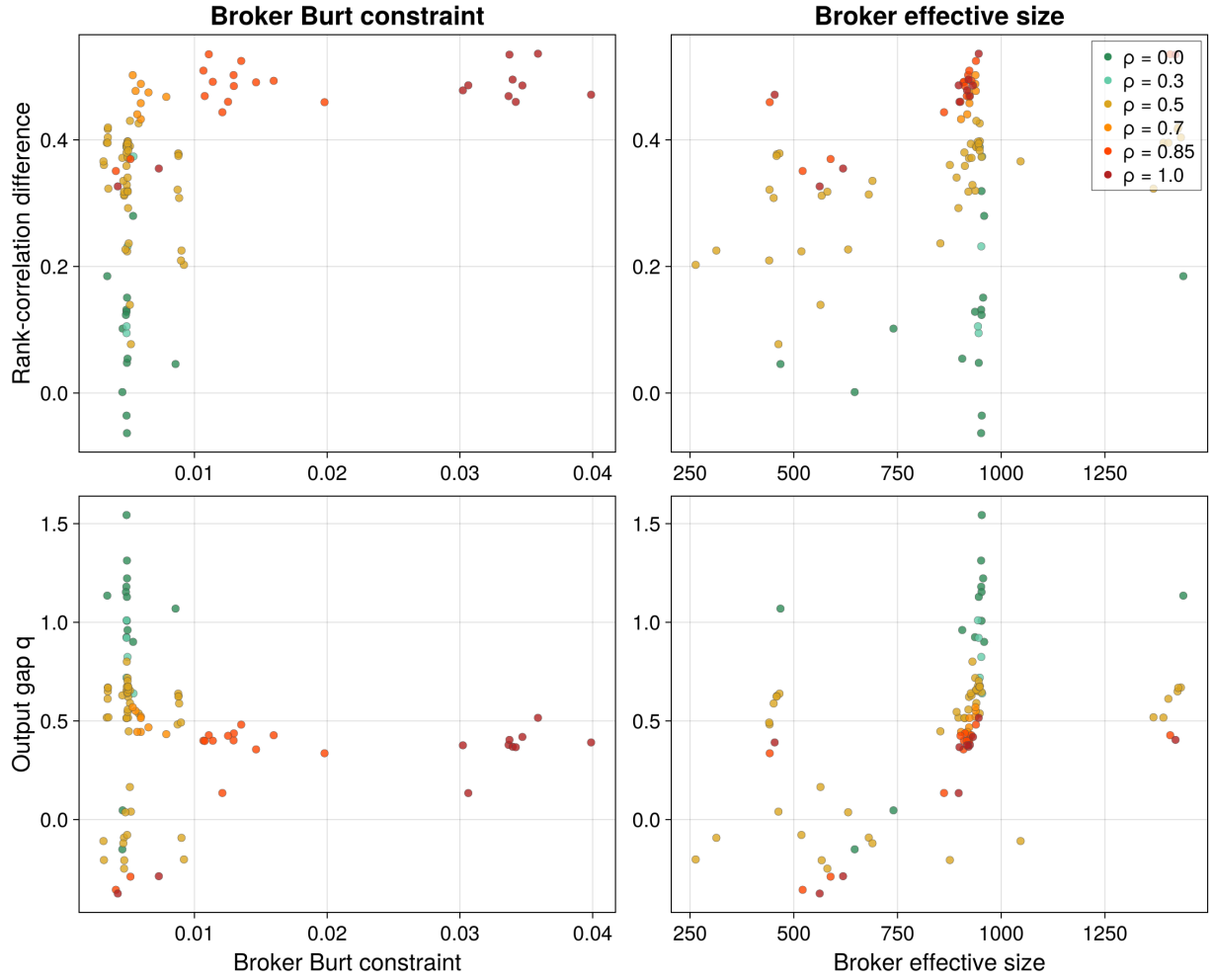


Figure S6: Difference in holdout rank correlation (broker minus principal, top) and output gap (broker channel minus self-search, bottom) against broker Burt constraint (left) and broker effective size (right). Each point is one regime's late-window mean across seeds, colored by ρ .